STATEMENT OF WORK (SOW)
Tactical Oceanography, Supporting Full-Spectrum Undersea Warfare

1.0 Background

The Naval Undersea Warfare Center is responsible for delivering undersea warfare (USW) capability that enables decisive warfighting advantage to the Fleet. The ocean environment impacts the performance and employment of USW systems and hence also directly informs the design of new systems. NUWC is expanding Tactical Oceanography (TAC-O) prototyping efforts that include the development of new onboard and offboard environmental sensors and devices that build operator situational awareness and information systems that enable operators to increase performance of existing systems through expert use of the environment.

The Navy is seeking the means to maintain definitive engagement superiority in the maritime environment in the face of global threats. The ability to exploit the environment to create increased performance margins for unmanned systems, weapons, fixed systems, sonar systems, communications and defensive systems is critical to maintaining USW dominance. Early TAC-O successes, driven by prototyping and experimentation, have created an increased demand signal. NUWC requires technical expertise and support in the areas of oceanography, ocean acoustics, non-acoustics, USW system simulation, experimentation, prototype sensor and device development, systems engineering, operations research and analysis, and submarines tactics. These technical expertise and services are required to support research and development efforts associated with improving USW with existing systems, informing the next generation attack submarine and enabling the hybrid Fleet. The TAC-O team will conduct research and development that spans from concept development through experimentation and prototypes fully integrated into Navy platforms and tested at sea.

Work associated with this task order includes Top Secret/Sensitive Compartmented Information (TS/SCI) and Alternate Compensatory Control Measures (ACCM). Operations research and analysis efforts involving TS/SCI information will be conducted at any of the authorized places of performance in Section 1.1 with appropriate facility clearance. Platform concept analyses using threat specific data, existing platform measured data for future capability workshops or wargaming efforts may also require clearance at the TS/SCI level and Alternative Compensatory Control Measures (ACCM access).

1.1 Places of Performance

- Agreement Holder Facilities
- Naval Undersea Warfare Center, Newport, RI
- Naval Undersea Warfare Center, Panama City, FL
- Naval Surface Warfare Center, Carderock, Bethesda, MD
- Naval Surface Warfare Center, Crane, IN
- Naval Surface Warfare Center, Philadelphia, PA
- Undersea Warfighting Development Center, Groton, CT

- Undersea Warfighting Development Center, San Diego, CA
- Commander Submarine Forces Pacific, Honolulu, HI
- U.S. Strategic Command, Omaha, NE
- Johns Hopkins Applied Physics Lab, Laurel, MD
- Naval Air Station Patuxent River, Patuxent, MD
- Seattle/Keyport, WA
- Washington, DC
- Other Government sites as specified in Technical Instructions

1.2 Authorized Users: NUWC NPT, NUWC KPT, CNMOC, NSWC PCD, NSWC CD

1.3 Sponsors: Office of Naval Research (ONR), OPNAV N975, PMS 351, PMS 401, PMS 450, PMS 392, PMS 394, NAVSEA 073, SSP, RDSA, IWS 5, PMW 770, PMS 406, PMS 404, DARPA, UWDC, COMSUBLANT, COMSUBPAC

1.4 Types of Funding: RDTE, O&MN, OPN, RDDA

2.0 Scope

The Agreement Holder shall provide scientific and engineering prototype deliverables in the areas of basic and applied meteorological and oceanographic (METOC) sciences, tactical systems analysis, acoustics and signal processing, and information systems design systems engineering, operations research, and operational analysis in support of the research and development efforts associated with Tactical Oceanography and the Full Spectrum Undersea Warfare Innovative Naval Prototype (FSUSWINP).

The Agreement Holder shall work with the Navy technical community to investigate research and development focus areas to ensure that the TAC-O prototype initiative can be scaled to serve the rapidly increasing demand. The Agreement Holder will contribute to the development of TAC-O program methodologies which may transition into existing acquisition programs of record for integrated submarine combat systems while the S&T efforts continue to develop the applied ocean sciences and a broader set of advanced capability prototypes.

3.0 Applicable Documents

The Agreement Holder shall perform the tasking required in Section 4.0 in accordance with the below Applicable Documents.

NUMBER	TITLE	REVISION/ DATE	APPLICABLE SOW TASK
3.1	MIL-STD-31000A, Department of Defense Standard Practice, Technical Data Packages	A	ALL
3.2	OPNAVINST 5513.5 SCG ID# 05-090, Submarine Technology for New Design Nuclear Powered Attack Submarines	N/A	ALL

	Commencing with SEAWOLF Class (SSN-21)		
3.3	MIL-STD-46855, Human Engineering Requirements for Military Systems, Equipment, and Facilities	A	ALL
3.4	DOD 5000.04-M-1 Agreement Holder Cost Data Reporting Manual	18 Apr 2007	ALL
3.5	MIL-STD-130N Identification Marking of U.S. Military Property (See http://www.assistdocs.com)	Rev. N	ALL
3.6	DOD 5220.22-M National Industrial Security Program Operating Manual (See http://www.dtic.mil/whs/directives/)	28 Feb 2006	ALL
3.7	FSUSWINP Project Specific Protection Plan	Sept 2023	ALL
3.8	DoD Program Manager's Guidebook for Integrating the Cybersecurity Risk Management Framework (RMF) into the System Acquisition Lifecycle	Sep 2015 Version 1.0	4.4
3.9	DoD Cybersecurity Test and Evaluation Guidebook	1 Jul 2015 Version 1.0	4.4
3.10	DoDI 8500.01 Cybersecurity (See http://www.dtic.mil/whs/directives/)	14 Mar 2014	4.4
3.11	DoDI 8510.01 Risk Management Framework (RMF) for DoD Information Technology (IT) (See http://www.dtic.mil/whs/directives/)	12 Mar 2014	4.4
3.12	NIST SP800-30, Guide for Conducting Risk Assessments (http://csrc.nist.gov/publications/PubsSPs.html#800-30)	23 Apr 2021	4.4
3.13	NAVSEA 9400.2-M NAVSEA PIT-Control System Cybersecurity Implementation Manual (NAVSEA Manual, Ser 05H/096, dtd 06 Oct 2016) (Authorization document: NAVSEA Instruction, Ser 00/365, dtd 20 Sep 2016, https://navsea.portal.navy.mil/hq/Docs/Instructions/09400.02A.pdf)	Oct 2016 Version 5	4.4
3.14	DoD Security Technical Implementation Guidance (STIGs) (DISA website: http://iase.disa.mil/stigs)	N/A	ALL
3.15	DoD Security Requirements Guides (SRGs) (DISA website: http://iase.disa.mil/stigs/)	N/A	ALL
3.16	IEEE STD 12207 Systems and Software Engineering - Software life Cycle Processes (See http://www.astm.org/standards)	Nov 2017	ALL

3.17	DoD 8570.01-M Information Assurance Workforce Improvement Program (http://dtic.mil/whs/directives/corres/pdf/857001m.pdf)	19 Dec 2005 Change 4 dated 10 Nov 2015	ALL
3.18	ANSI/EIA-649-A National Consensus Standard for Configuration Management (See http://webstore.ansi.org/)	Oct 2004	ALL
3.19	SAE AS5553B Counterfeit Electrical, Electronic, and Electromechanical (EEE) Parts; Avoidance, Detection, Mitigation, and Disposition	12 Sep 2016	ALL
3.20	SAE AS6462A Fraudulent/Counterfeit Electronic Parts; Avoidance, Detection, Mitigation, and Disposition Verification Criteria	26 Aug 2014	ALL
3.21	SAE AS6081 Fraudulent/Counterfeit Electronic Parts: Avoidance, Detection, Mitigation, and Disposition – Distributors	07 Nov 2012	ALL
3.22	SAE ARP6178 Fraudulent/Counterfeit Electronic Parts; Tool for Risk Assessment of Distributors	19 Dec 2011	ALL
3.23	SAE AS6174 Counterfeit Materiel; Assuring Acquisition of Authentic and Conforming Materiel	29 Jul 2014	ALL
3.24	SAE AS6301 Compliance Verification Criterion Standard for SAE AS6081, Fraudulent/Counterfeit Electronic Parts: Avoidance, Detection, Mitigation, & Disposition-Dist	14 Jan 2014	ALL
3.25	SECNAV M-5239.2, Cyberspace Information Technology And Cybersecurity Workforce Management And Qualification Manual, (As amended 20 May 2020 by DoN CIO Memorandum) June	June 2016	6
3.26	DoD Directive 8140.01, Cyberspace Workforce Management	05 Oct 2020	4.6
3.27	DoD Instruction 8140.02, Identification, Tracking, And Reporting Of Cyberspace Workforce Requirements	21 Dec 2021	4.6
3.28	DoD Manual 8140.03, Cyberspace Workforce Qualification And Management Program	15 Feb 2023	4.6

4.0 Technical Requirements

Below enumerated tasking is a mix of direct prototype creation and analysis work. The analysis work is in service of identifying additional capability that will result in prototype deliverables that incorporate that additional capability as long as funding is available and allocated to support such incorporation.

4.1 Concept Development, Prototyping, and Analysis

4.1.1 The Agreement Holder shall develop prototype concepts for specific USW systems (onboard and offboard) and related interfaces required to support the employment of various future offensive and defensive system payloads based on oceanographic performance sensitivities and projected capabilities derived from technical exchange events or other Navy source data provided as GFI. The Agreement Holder shall analyze data developed through mission evaluation modeling and simulation capabilities from a submarine operational and tactical position, unmanned systems, weapons, and fixed systems. This includes consideration of oceanographic parameters, sensitivity, and sensor impacts on USW system performance and employment. USW systems include: submarine sonar, weapons, fixed and mobile surveillance systems, unmanned systems, communications systems and countermeasures.

4.1.2 The Agreement Holder shall develop concepts and deliver concept prototypes, define employment tactics, tactical rulesets, and ruleset test cases in support of technical insertion capabilities.

4.1.3 The Agreement Holder shall provide human systems integration and human factors engineering analysis support for the integration of future capabilities within the relevant design space. This includes developing prototype operator interfaces and decision aids for oceanographic sensor data.

4.1.4 The Agreement Holder shall provide systems engineering services to perform and deliver technical studies to support the evaluation of new technologies being considered for implementation into both the launch platforms systems, as well as the payloads being employed by interfacing to these systems. This includes the evaluation of new oceanographic sensor technologies for integration into launch platforms and payloads.

4.1.5 The Agreement Holder shall provide systems engineering, innovation engineering, and subject matter expertise in events, including concept or technology brainstorming sessions, wargaming exercises, or other submarine-related tactical evaluation and assessment workshops. The contactor shall prepare, host, and facilitate these events to be attended by Government and Agreement Holder personnel. This includes addressing oceanographic considerations in system-of-systems capabilities.

4.1.6 The Agreement Holder shall gather USW platform information to support engineering modifications, engineering drawings, and provide expertise to assemble and deliver Technical Data Packages (TDPs) in support of Temporary Alteration (TEMPALT) of ship.

4.1.7 The Agreement Holder shall modify or develop new software at Collateral Classification or TS/SCI Security Level to support data collection and implement of software for TEMPALTS associated with 4.1.6

4.1.8 The Agreement Holder shall assemble IT artifacts, implement IT controls and firmware, and documentation to gain Interim Authority to Test (IATT), Local Authority to Test (LATTE) and Authority to Operate (ATO) for both ashore and afloat based systems.

4.1.9 The Agreement Holder shall conduct analysis including modeling and simulation of new concept prototypes and capabilities and associated employment tactics. The Agreement Holder shall support live, virtual, constructive events using the modeling and simulation tools. This includes M&S to ensure oceanographic sensor placement and performance will meet requirements to enhance performance of both on-platform and deployed systems and sensors.

4.1.10. The Agreement Holder shall conduct analysis to inform requirements for marine robotics sampling of the ocean environment and impact of marine robotics and environment on subsea seabed warfare systems.

4.1.11 The Agreement Holder shall provide preliminary systems engineering support, including Model Based System Engineering, for the concepts and capabilities developed under 4.1.9 and 4.1.10 above. This includes modeling and simulation of oceanographic sensor data.

4.2. Analysis of Future Systems Operation

4.2.1. The Agreement Holder shall evaluate capability concepts using analytical or modeling and simulation tools, along with submarine operationally experienced personnel to assess submarine operational performance and employment tactics. The Agreement Holder shall deliver a report documenting its assessment of the specific concepts. This includes assessing the impact of oceanographic conditions on submarine operational performance and employment tactics, including sensor performance.

4.2.2 The Agreement Holder shall conduct data analysis using GFI or Agreement Holder generated modeling and simulation data, current generation systems analogous data and other relevant data sources in order to evaluate the potential performance gains and impacts on a submarine's ability to complete the required missions. This includes conducting digital signal processing, developing algorithms and displays of oceanographic sensor data to evaluate performance gains.

4.2.3 The Agreement Holder shall conduct feasibility studies and support Analysis of Alternatives, in accordance with applicable performance specifications, encompassing platform combat systems and subsystems and considering platform missions and potential operating profiles. These studies shall assess risk and options to mitigate as well as assessing potential costs, required resources and impact on platform reliability, maintainability and availability. This includes analyzing the feasibility of integrating ruggedized oceanographic sensors suitable for use and deployment from submarines, and conducting stress, vibrational, shock and other testing and analysis to support ship alteration packages.

Additional Deliverables (Standalone Section or Integrated into Existing Deliverable Lists): The Agreement Holder will provide the following deliverables:
- Prototype Algorithms for processing oceanographic sensor and USW system data.
- Feasibility Studies regarding the integration of oceanographic sensors and their impact on USW systems to identify potential prototypes.
- Oceanographic Sensor and USW System Analysis Reports detailing sensor performance, sensitivity, and impact on USW system effectiveness in support of potential and delivered prototypes.
- Ruggedized Oceanographic Sensors suitable for use and deployment from submarines, marine robotics, weapons, and other USW infrastructure (hardware and associated documentation). This includes the design, attachment, processing and evaluation of submarine oceanographic sensors directly on platform and those launched offboard.
- Prototype Operator Interfaces and Decision Aids for displaying and interpreting oceanographic sensor data.

5.0 Progress Reports

Progress reporting will be identified for each prototype project.

6.0 Government Furnished Information

Government Furnished Information (GFI) will be made available for each prototype project, as needed.

7.0 Milestone Payment Schedule

Milestone payments work very similarly to performance based payments. Milestone events will be determined for each prototype project.

8.0 Security Classification

The required security level for the prototype project will be Top Secret/Sensitive Compartmented Information (TS/SCI).

8.1 Program Protection

SECURITY: All Agreement Holder personnel shall adhere to the Security provisions of 32 CFR Part 117 – National Industrial Security Program Operating Manual (NISPOM) and the security requirements of the attached DD254 (if included). While performing work at a Government

Facility, Agreement Holder personnel shall comply with the security regulations of the host facility. Applicable FAR, DFARS, NMCARS clauses, and NAVSEA text shall be adhered to in the performance of this agreement. Security incidents shall be promptly reported through the company's Facility Security Officer (FSO), to the Agreement Officer's Representative (AOR), Technical Point of Contact (TPOC), and the Cognizant Security Office to NUWCDIVNPT Security.

Controlled Unclassified Information (CUI) including Legacy FOUO and Covered Defense Information (meeting the definition of 48 CFR 252.204–7012(a)) generated and/or provided under this agreement shall be marked and safeguarded as specified in DoD Instruction 5200.48, CUI available at:
https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/520048p.PDF. Any product containing Covered Defense Information shall be assigned a distribution statement (distribution statements B through F) in accordance with DoDI 5230.24 (Distribution Statements on Technical Documents); and DoDI 5230.24, Enclosure 3 Procedures, available at
https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/523024p.pdf

INFORMATION SECURITY: If the work is performed at the Agreement Holder's facility, the Agreement Holder shall implement and maintain security procedures and controls to prevent unauthorized disclosure of classified information and controlled unclassified information (CUI) and to control distribution of CUI in accordance with National Industrial Security Program Operating Manual (NISPOM) codifying 32 CFR Part 117, NISPOM Rule, and SECNAV M-5510.36B. If the work is performed at the Government's facility, the Agreement Holder shall comply with facility policy.

CUI INCIDENT REPORTING AND RESPONSE: The Agreement Holder shall promptly report any unauthorized, inadvertent, or illegal release or disclosure of CUI to the AOR / TPOC, Agreements Officer, and the Security Office. Agreement Holder personnel shall coordinate this effort through the relevant industry site FSO.

PUBLIC RELEASE: Any controlled unclassified information pertaining to this agreement shall not be released for public dissemination, including posting to any social media sites such as Facebook or Twitter, unless it has been approved for public release by appropriate U.S. Government authority. Proposed public releases shall be submitted for approval prior to release through the appropriate U.S. Government Office.

8.2 Operations Security (OPSEC)

OPSEC is a process that identifies critical information to determine if friendly actions can be observed by adversary intelligence systems, determines if information obtained by adversaries could be interpreted to be useful to them, and then executes selected measures that eliminate or reduce adversary exploitation of friendly critical information.

The Agreement Holder shall develop and implement, and update and maintain an OPSEC program to protect controlled unclassified and classified activities, information, equipment, and material used or developed by the Agreement Holder and any subagreement holder during performance of the agreement. The Agreement Holder shall be responsible for the subagreement holder implementation of the OPSEC requirements. The Agreement Holder developed OPSEC program may include Information Assurance and Communications Security (COMSEC). The OPSEC program shall be in accordance with National Security Presidential Memorandum (NSPM) 28, and at a minimum shall include:

1) Assignment of responsibility for OPSEC direction and implementation.
2) Issuance of procedures and planning guidance for the use of OPSEC techniques to identify vulnerabilities and apply applicable countermeasures.
3) Establishment of OPSEC education and awareness training.
4) Provisions for management, annual review, and evaluation of OPSEC programs.
5) Flow down of OPSEC requirements to subagreement holders when applicable.

While performing aboard Government sites, the Agreement Holder shall: comply with all OPSEC instructions and policies; include OPSEC as part of its ongoing security awareness program and take all required Agency training; Be responsive to the Supporting OPSEC Manager on a non-interference basis; and Protect sensitive unclassified information and activities, which could compromise classified information or operations, or degrade the planning and execution of operations performed by the Requiring Organization and Agreement Holder in support of the mission.

8.3 Additional Requirements

<u>Sensitive Compartmented Information (SCI)</u>
SCI is required for the Agreement Holder to perform task(s) 4.1 and 4.2.

<u>Non-Sensitive Compartmented Information (non-SCI)</u>
Non-SCI is required for the Agreement Holder to perform task(s) 4.1 and 4.2. The Agreement Holder will require access to non-SCI Intelligence information when accessing SCI information.

<u>NATO</u>
NATO is required for the Agreement Holder to perform task(s) 4.1 and 4.2. The Agreement Holder will require NATO access when working on IT systems within the SCI realm.

<u>Alternative Compensatory Control Measures (ACCM) Information</u>
ACCM is required for the Agreement Holder to perform task(s) 4.1 and 4.2. The tasks may involve having the Agreement Holder directly handle ACCM data and ACCM products that are derived from the ACCM data.

<u>Naval Nuclear Propulsion Information (NNPI)</u>
NNPI (Classified and Unclassified) is required for the Agreement Holder to perform task(s) 4.1 and 4.2. The tasks may involve having the Agreement Holder directly handle NNPI data and NNPI products that are derived from the NNPI data.

NNPI is controlled and protected under one or more of the following: (1) the Atomic Energy Act of 1954 (as amended), (2) the 1984 Defense Authorization Act, (3) Export Control Act

Regulations of the Commerce Department, (4) Arms Export Control Act Regulations (i.e., International Traffic in Arms Regulations (ITAR); Munitions List), (5) Export of Sensitive Nuclear Information for Foreign Atomic Energy Regulations of the Energy Department, and Safeguarding of Naval Nuclear Propulsion Information (OPNAVINST N9210.3).

a. NNPI is information related to sensitive military technology (i.e., naval nuclear propulsion technology).
b. Access to NNPI is limited to U.S. citizens with a need to know.
c. A final clearance is required for access to NNPI.
d. NNPI is marked and handled as NOFORN.